RAKESH CHOWDARY POLINA

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PROFILE

Experienced Software Engineer with a proven track record in developing scalable microservices, implementing robust facial recognition pipelines, and delivering cloud-native solutions. Skilled in Python, Java, Spring Boot, Flask, React.js, and AWS, with expertise in containerization, CI/CD, and Agile methodologies. Adept at optimizing performance, enhancing user engagement, and driving operational efficiency in production environments.

EDUCATION

Master of Science in Computer Science **May 2024**
Oklahoma State University, Stillwater, OK **CGPA: 3.909/4**

SKILLS

Languages:	Java, JavaScript, Python, SQL, C, C++, HTML, CSS, Swift
Frameworks:	React JS, Angular JS, Spring boot, Node JS, Hibernate, Spring MVC, React Native
Databases:	MySQL, MongoDB, NoSQL, PostgreSQL, Cassandra
Web Technologies:	RESTful API, Spring JDBC, JPA, OAuth 2, jQuery, JSP, JMS, RabbitMQ
Web Services:	REST, SOAP
Tools/Platforms:	Docker, Git, JIRA, Jenkins, Postman, Ansible, Kubernetes, Apache Kafka, Terraform, Swagger
AWS Cloud Services:	EC2, S3, EKS, RDS, ELB, Cloud Watch, CloudFront
Build Tools:	Maven, Gradle

WORK EXPERIENCE

Research Data Analyst, Oklahoma State University **May 2023 – May 2024**

- Engineered and implemented a robust facial recognition pipeline, performing both quantitative and qualitative analysis on image data with Python and InsightFace, to achieve a precision rate of 93.6%
- Implemented DBSCAN clustering and rigorously verified data accuracy across 182,000+ images, enhancing archive reliability and insight precision.
- Designed an intuitive UI for the labeling tool using Flask and React.js, streamlining user engagement and annotations.
- Designed and maintained database structures using SQL, enhancing user interaction management and feedback data integrity, ensuring operational excellence.

Software Engineer, Ericsson, India **July 2019 – July 2022**

- Analyzed monolithic architecture, transitioning to microservices with RESTful APIs using Java and Spring Boot. Integrated Apache Kafka for asynchronous communication, improving system scalability by handling 50% more concurrent users. Enhanced operational efficiency by 20% through effective service decomposition and asynchronous communication patterns.
- Performed comprehensive requirement analysis and crafted technical specifications for API development. Reduced system downtime by 30% by implementing Spring Security and Spring Data for robust transaction management and secure data access. Optimized resource utilization in Oracle database management to support complex ERP-level workflows.
- Designed and implemented robust APIs using Spring Boot, adhering to industry standards. Utilized Spring WebFlux for building reactive, non-blocking APIs, reducing API response time by 40%. Incorporated Spring MVC for building interactive web applications, enhancing user interface responsiveness and scalability. Developed stored procedures to handle data processing efficiently, ensuring seamless integration with backend systems.
- Executed Swagger for precise API documentation and validation, leading to a 40% reduction in onboarding time for new developers. Ensured robust unit testing and API simulation using Postman, improving test coverage and endpoint reliability. Maintained application reliability in Workday Studio and Workday Extend implementations.
- Created and modified Oracle PL/SQL packages, procedures, triggers, and functions to handle complex data workflows and optimize backend operations. Utilized Oracle SQL for seamless data retrieval and transformation. Supported database structures within ERP-level systems, employing backup methodologies and monitoring tools to ensure reliability.
- Developed dynamic and responsive user interfaces using ReactJS, improving user engagement. Incorporated Tableau and IBM Cognos Analytics for data visualization and reporting, enabling actionable insights for business stakeholders. Supported higher education systems through Workday Tenant Management, ensuring configurations and data integrity.
- Implemented Docker and Kubernetes for containerization and orchestration, cutting deployment time by 50%. Utilized Helm for managing Kubernetes deployments and automating infrastructure configurations.
- Streamlined CI/CD pipelines using Jenkins, enhancing the software development lifecycle efficiency. Reduced time-to-market by 40% and minimized production incidents by 20% through automated testing and deployment strategies. Managed Linux server administration to support application hosting and infrastructure stability.
- Followed AGILE (SCRUM) methodologies with bi-weekly sprint planning and daily status meetings to monitor project progress and ensure alignment with product goals. Collaborated with cross-functional teams, ensuring seamless communication and coordination. Addressed stakeholder concerns and provided effective solutions in higher education environments.